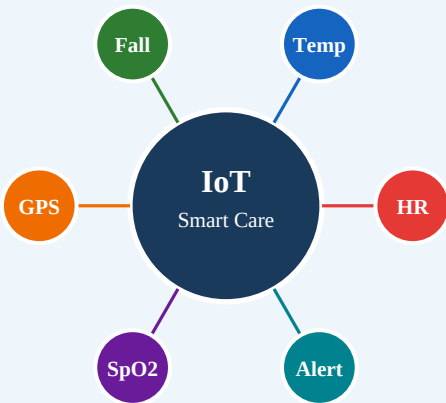


IoT-Based Smart Elderly Care Monitoring System

A Research Report — CodeAlpha IoT Internship

IoT Sensor Network Overview



Internship Submission Details

Submitted by	Priyadharshini K
Department	Electronics and Communication Engineering
College	Mailam Engineering College
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1. Abstract

IoT (Internet of Things) technology is growing very fast and it is now being used in many areas including healthcare. In this report, I have studied about an IoT-Based Smart Elderly Care Monitoring System. The main idea of this system is to keep track of old age people health in real time using small sensors that they can wear. The system uses a heart rate sensor, temperature sensor, and a fall detection sensor. All the data from these sensors goes to an ESP32 microcontroller and then it is sent to a cloud database through Wi-Fi. When something is wrong, the caregiver or family member gets an alert on their mobile phone immediately. This report helped me understand how IoT can be used to solve real-life problems in healthcare.

2. Introduction

These days, the number of old age people is increasing a lot. Many of them live alone at home or in old age homes. The main problem is that nobody is there to watch them all the time. If they fall down or if their health becomes bad suddenly, there is nobody to help them quickly. Because of this reason, many old people face serious health problems that could have been avoided if someone noticed it early.

IoT technology gives us a good solution for this problem. Using IoT, we can connect small sensors to the internet and monitor the health of elderly people from anywhere. During this internship, I studied how an IoT-based elderly care monitoring system can be developed using simple components such as ESP32, sensors, and a mobile application. This helped me understand how connected devices can support healthcare monitoring. The goal of this project is to make sure old age people are safe even when no one is physically present near them.

3. Problem Statement

Old age people who live alone face many health related dangers. They can fall down, get fever, or have heart problems at any time. The main issue is that these things

happen suddenly and there is no one to notice it. Traditional methods like checking on them once a day or having a nurse visit weekly is not enough. If a person falls down and becomes unconscious, they cannot press any emergency button to call for help.

Also, most of the smart wearable devices available in the market are very costly and complicated to use. There is no simple, affordable system that can automatically detect health problems and send alerts to family members without needing the elderly person to do anything. This is the main gap that I wanted to address in this project.

4. Objectives

1. To study how wearable health monitoring systems work for elderly care.
2. To understand how ESP32 or Arduino can be used in IoT applications.
3. To learn how cloud platforms can store and monitor health data.
4. To understand how alert notifications can help caregivers.
5. To explore practical applications and future possibilities of IoT in healthcare.

5. System Architecture

The system is divided into four main parts. First is the sensor layer where all the physical sensors are connected to the elderly person body. Second is the controller layer where ESP32 collects all the sensor data. Third is the cloud layer where the data is stored and processed. Fourth is the application layer where the caregiver can see the data and receive alerts on their phone. Each part is connected to the next one and the whole system works automatically without needing manual operation.

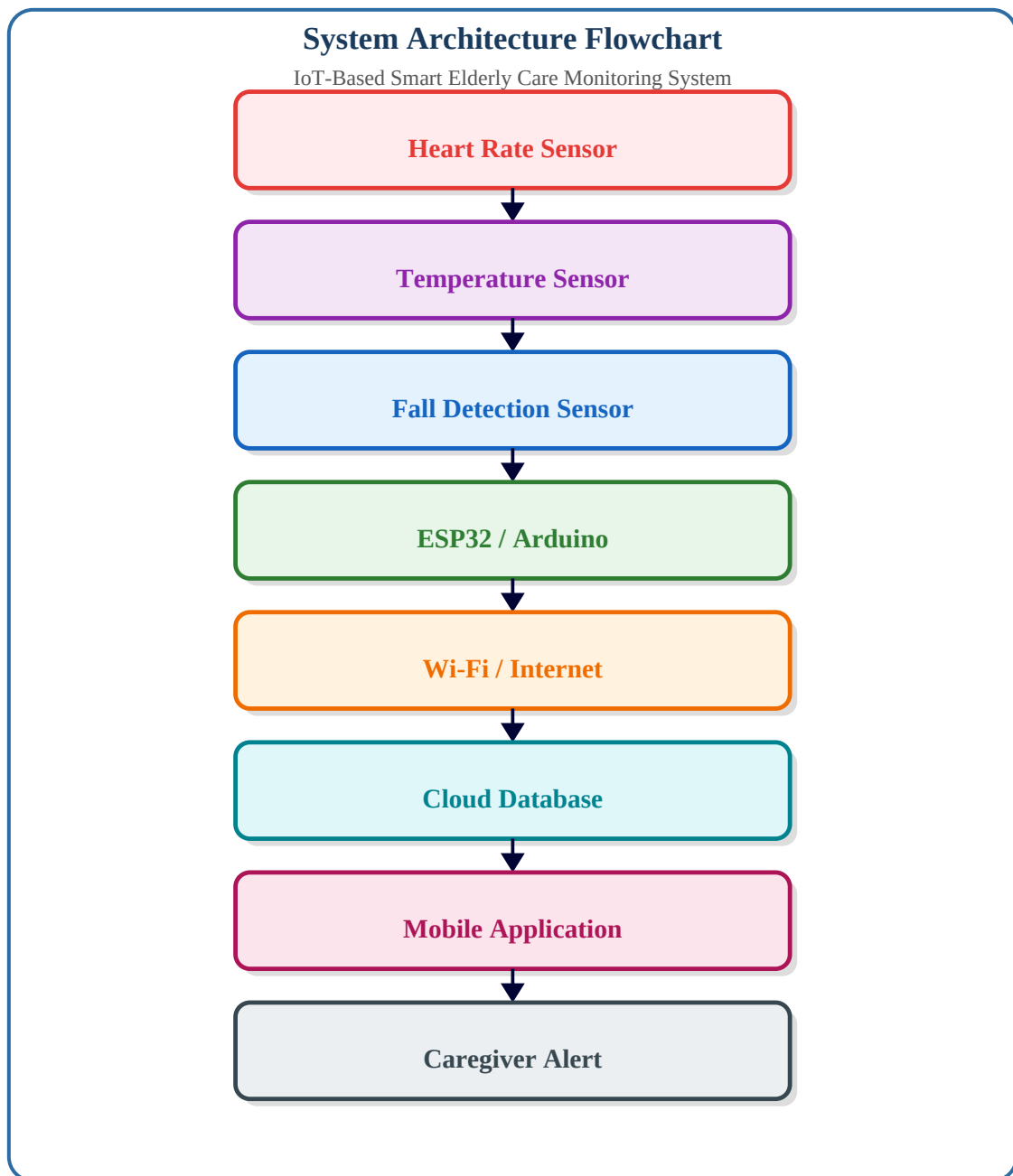


Figure 1: System Architecture Flowchart — IoT-Based Smart Elderly Care Monitoring System

The heart rate sensor MAX30102 is used to measure the pulse rate. The DS18B20 sensor is used to measure body temperature. The ADXL345 accelerometer is used to detect if the person has fallen down. All these sensors are connected to the ESP32 board. The ESP32 processes the data and sends it to the cloud server using Wi-Fi and MQTT protocol.

6. Working Principle

The system works continuously in the background. The heart rate is measured every 10 seconds, temperature is checked every 60 seconds, and the accelerometer checks for falls at a rate of 50 times per second. The ESP32 reads all this data and checks if the values are within safe limits. If everything is normal, the data is simply stored in the cloud. But if any value crosses the safe limit, an alert is sent immediately to the caregiver mobile app.

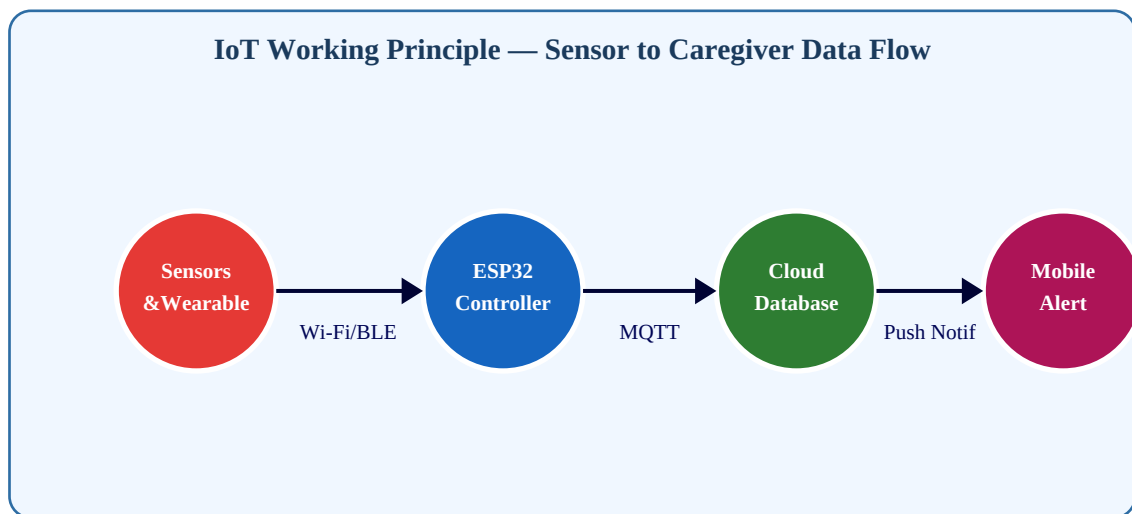


Figure 2: IoT Working Principle — Data Flow from Sensors to Caregiver Alert

For example, if the heart rate goes below 45 or above 130, the app sends a heart alert. If the temperature goes above 38.5 degree Celsius, a fever alert is sent. If the accelerometer detects a sudden fall, an SOS message is sent to the caregiver within 2 seconds. The mobile app also shows a live graph of the health data so the caregiver can check the person condition at any time.

7. Applications

This system can be used in many places where elderly people need to be monitored. Some of the main applications are listed below:

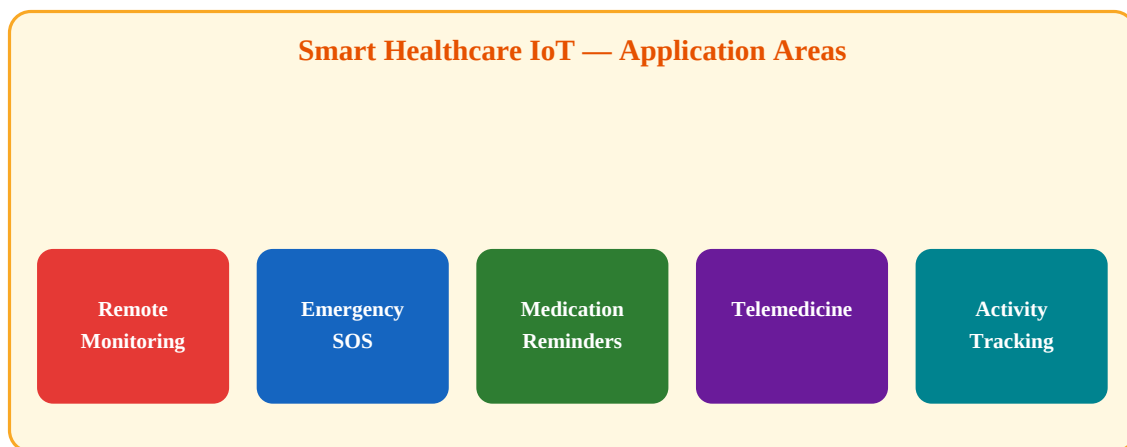


Figure 3: Smart Healthcare IoT — Key Application Areas

- **Home Monitoring:** Old age people living alone at home can be monitored by their family members from a different location using this system.
- **Old Age Homes:** In old age homes, the staff can monitor many patients at the same time using one central dashboard connected to this system.
- **After Hospital Discharge:** Patients who recently came out of hospital can be monitored at home to make sure they are recovering properly.
- **Chronic Disease Patients:** People who have long term diseases like diabetes or heart problems can use this system for daily health tracking.

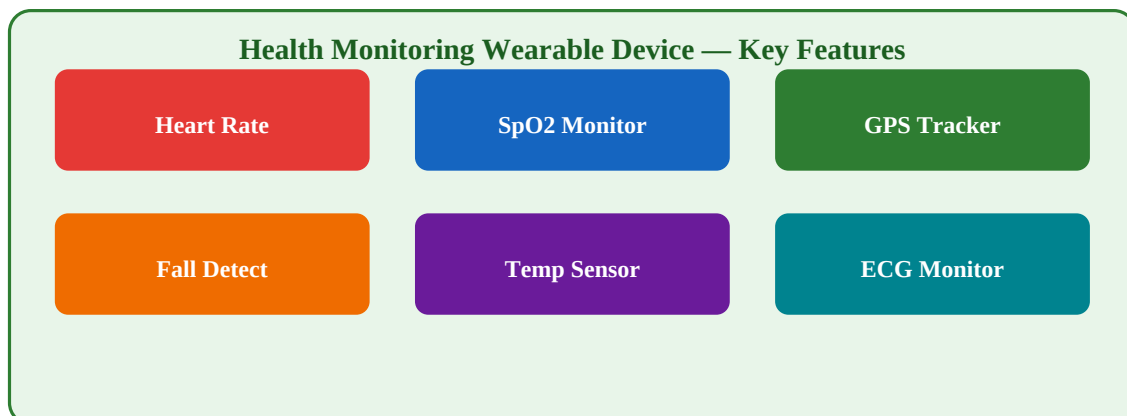


Figure 4: Health Monitoring Wearable Device — Feature Overview

8. Advantages

- + **Real-Time Alerts:** The system sends alerts within seconds when a health problem is detected. So the caregiver can take action immediately.
- + **Easy to Wear:** The sensors are small and light. The elderly person can wear them without any difficulty in their daily activities.
- + **Low Cost:** Compared to other hospital monitoring systems, this IoT based system is much cheaper and affordable for common people.
- + **Works from Anywhere:** Since the data is stored in the cloud, the family members can check the health status from anywhere using just a mobile phone.
- + **Health History:** All the health data is saved in the cloud so doctors can also look at previous records when needed.
- + **Less Burden on Family:** Family members do not need to be physically present all the time. The system keeps them informed automatically.

9. Challenges and Limitations

Even though this system has many benefits, there are also some problems that I found during my study. These challenges need to be solved before this system can be used widely.

- ! **Internet Dependency:** The system needs a stable internet connection all the time. If the Wi-Fi goes off, the data will not reach the cloud and alerts will not be sent.
- ! **Battery Life:** The sensors and ESP32 need power continuously. The battery of the wearable device will drain fast if not managed properly.
- ! **Data Privacy:** The health data of the patient is sensitive. It should be properly secured so that no unauthorized person can access it.
- ! **Sensor Errors:** Sometimes the sensors can give wrong readings due to body movement or sweat. This may cause false alerts.

! Old People Hesitate to Use Technology: Some elderly people are not comfortable using smart devices. It takes time and effort to train them to use this system.

10. Real-World Case Study

As part of this study, I explored an example scenario of how an IoT-based elderly monitoring system may work in real life. In this scenario, elderly people wear a small health monitoring device containing sensors for heart rate, temperature, and fall detection. The sensor data is sent to the cloud through an ESP32 module and can be viewed through a mobile application.

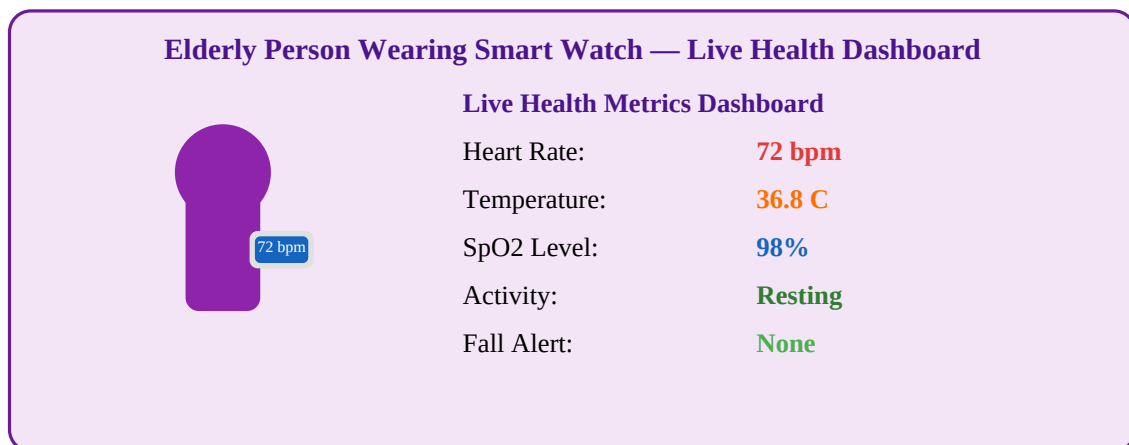


Figure 5: Elderly Person Wearing Smart Watch — Live Health Metrics Dashboard

If an unusual condition is detected, such as abnormal heart rate, high temperature, or a fall, the system sends an alert to the caregiver. This type of monitoring can help family members respond quickly and improve elderly safety. This example helped me understand how IoT technologies can support healthcare and make remote monitoring more practical.

11. Future Scope

There are many things that can be added to this system in the future to make it even better. Some ideas that I thought of are:

- We can add machine learning to the system so that it can predict health problems before they actually happen, not just detect them after.
- More sensors like ECG for heart monitoring and SpO2 for blood oxygen level can be added to the wearable device.
- Instead of sending all data to cloud, some processing can be done inside the ESP32 itself to reduce internet dependency.
- A voice call feature can be added so that the system automatically calls the caregiver during an emergency.
- The system can be connected to smart home devices like automatic lights and door locks so the environment also becomes safer for elderly people.
- To protect patient data, blockchain technology can be used for secure storage in the future.

12. Conclusion

In this report I studied and explained an IoT-Based Smart Elderly Care Monitoring System in detail. The main goal of this system is to keep old age people safe by monitoring their health continuously using simple sensors and sending alerts to their family members when needed.

From studying this topic, I understood how IoT technologies can improve elderly healthcare through continuous monitoring and alert systems. This report helped me learn about sensors, cloud communication, wearable monitoring, and the practical use of IoT in healthcare applications.

I believe technologies like this can become useful in supporting elderly people in the future while reducing caregiver workload. Preparing this report as part of my

CodeAlpha Internship helped me understand how IoT systems are designed and how they can support real-life healthcare applications.

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